

Cristian Francesco Pennino

Software Engineer • Robotics | Computer Vision | Embedded Systems
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SUMMARY

Software engineer specializing in robotics, computer vision and embedded systems, with full-stack development in React, TypeScript and PostgreSQL and low-level firmware in C/C++ on STM32. Strong foundation in data structures, algorithms and Linux, proven through international competition. Seeking early-career roles in robotics, embedded or backend software engineering.

COMPETITIONS & AWARDS

- **RoboCup 2026 World Championship** - 3rd place, Rescue Simulation (Incheon, South Korea); **1st place national 2026**.
- **STMicronics STM32ODE** - 3rd place national (2024, "Build the Future with STM32").
- **OliCyber.IT** (national cybersecurity olympiad) - national finalist 2024 and 2026 (top 100 of 3,200+ participants).
- **CyberChallenge.IT** - selected participant, University of Catania (ethical hacking, cryptography, CTF).
- **Italian Informatics Olympiad (OI/OIS)** - competitor, individual and team.
- **Reply Code Challenge** - participant (team competitive programming).
- **NXP Summer School 2025** - intensive program in embedded systems and firmware development.

EDUCATION & CERTIFICATIONS

Diploma in Computer Science & Telecommunications (Perito Informatico) - 100/100 cum laude 2026
I.T.I.S. "E. Fermi", Giarre (CT), Italy - software development, systems & networks, databases, cybersecurity

Certifications: Cisco CCNA - Introduction to Networks (2025); Switching, Routing & Wireless Essentials (2026); Enterprise Networking, Security & Automation (2026). Cisco IT Essentials (2024). Cambridge English B2 First (2025).

SELECTED PROJECTS

RoboCupJunior Rescue Simulation - Perception Lead 2026
Python, OpenCV, Ultralytics YOLO, NumPy, Webots/Erebus 3rd World | 1st National

- Placed **3rd at the RoboCup 2026 World Championship** (Incheon, South Korea) and **1st nationally** (Italy) with team Black Radiators.
- Owned the computer-vision perception subsystem of a team-built autonomous rescue robot: HSV-based victim and hazard-sign detection with bounding-box localization.
- Curated and labeled a training dataset from simulation captures and trained a YOLO model for victim and hazmat-sign classification.
- Ran YOLO inference in the real-time control loop with asynchronous queuing to avoid latency spikes.
- Added hazmat scanline detection and a LiDAR-based verification step to reject false-positive ("fake letter") signs, improving map-scoring accuracy.

TechStore - Full-Stack E-Commerce Platform (solo) 2025
React, Vite, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Stripe

- Engineered an end-to-end e-commerce system: dynamic product catalogue, multi-criteria filtering, and a responsive checkout flow.
- Integrated Stripe Checkout server-side via Supabase Edge Functions; persisted orders to PostgreSQL and dispatched automated confirmation emails.
- Built an admin dashboard for product CRUD, inventory and order management with sales analytics (Recharts); enforced access with PostgreSQL Row Level Security.

AuthLog - Secure IoT Access-Control System (solo)

2025-2026

STM32, ESP8266, React Native, Supabase, TLS 1.2, Embedded C

- Designed a smart access-control system unifying embedded hardware, cloud and mobile: STM32 handles RFID/NFC acquisition; ESP8266 communicates with a Supabase/PostgreSQL backend over HTTPS (TLS 1.2).
- Implemented REST-over-TLS authentication with automatic logging of every access event to the backend, plus Time-of-Flight proximity detection for energy efficiency; identified and remediated a hardcoded-credential issue in the codebase.
- Built a cross-platform React Native app for user enrolment and direct NFC UUID writing from a smartphone, removing the need for physical badges.

ForestGuard - Distributed Solar LoRa Sensor Mesh (solo)

2024-2025

STM32WL55, Embedded C, LoRa 868 MHz, SPI/I²C/UART, sensor fusion, backend integration

- Built a solar-powered, distributed LoRa mesh of STM32WL55 nodes for forest-fire and intrusion detection.
- Combined multi-sensor acquisition with adaptive thresholding to minimize false positives, feeding a backend for centralized monitoring.

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript/JavaScript, SQL

Robotics & Computer Vision: OpenCV, YOLO (Ultralytics), NumPy, Webots/Erebus

Embedded & IoT: STM32 (F4/WL55), ESP8266, Raspberry Pi, Arduino, bare-metal C, SPI/I²C/UART, LoRa 868 MHz, RFID/NFC

Web & Backend: React, Next.js, Node.js, Supabase, PostgreSQL, Stripe, REST APIs

Systems & Networking: Linux, Git, Docker, CI/CD (GitHub Actions), TCP/IP, routing & switching, VLAN, NAT, DHCP, TLS

Security: OWASP Top 10, web exploitation (SQLi, XSS), cryptography, CTF, Burp Suite, Nmap, Wireshark, Ghidra

Foundations: data structures & algorithms, OOP, relational data modelling, software architecture, unit testing, Agile

Spoken languages: Italian (native), English (C1 listening / B2 Cambridge), Spanish (basic)